

Adaptive Evidence Fusion and Temporal Alignment for Multimodal Video Retrieval

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Abstract— We present an interactive multimodal video-retrieval system for Known Item Search (KIS) and Temporal Retrieval and Alignment of Key Events (TRAKE). A C++/FFmpeg sampler extracts a canonical 1-FPS frame corpus, which specialist models enrich with scene context, visible text, objects, speech, and speakers. Qwen3-VL, Florence-2, YOLOE, Qwen3-ASR, and pyannote produce this evidence; SigLIP2 embeds images and visual-language queries, and BGE-M3 embeds serialized FRAMECONTEXT and speech segments. Exact inner-product FAISS indexes store L2-normalized visual, context, and ASR vectors. A fielded BM25 index retains literal title, caption, OCR, and ASR matches. Detached frame search combines returned lists with weighted reciprocal-rank fusion. KIS and TRAKE instead score each event against the full corpus, calibrate and fuse available evidence at each frame, then decode strict chronological paths with a linear-time monotonic dynamic program. The React client provides search, evidence inspection, frame-accurate video navigation, history replay, and a revision-aware shared submission workspace. Offline GPU jobs and optional inference services run on disposable Thunder Compute instances; canonical metadata, indexes, and collaborative state remain local or in S3.

1. System Overview

MLeCDanBGGold anchors every layer to a canonical frame corpus (Appendix Fig. 1). Offline preparation extracts one frame per second, adds specialist evidence, builds dense and sparse indexes, validates identity and coverage, and publishes immutable bundles. The online backend opens these artifacts once, then handles retrieval, temporal decoding, canonical materialization, filtering, media resolution, and operator state. The React application handles inspection and submission; embeddings, corpus identity, ranking, and competition-coordinate conversion remain in the backend.

The corpus layer owns frame and evidence authority. Modality retrievers own their encoders and index-native search; evidence scorers construct calibrated event-frame matrices; the temporal package decodes numerical paths; and KIS/TRAKE workflows materialize task-specific responses. Canonical resolvers reject decoder positions absent from the corpus and prevent model adapters from rewriting `frame_idx`. A missing optional modality reduces the reported capability and never triggers an implicit online rebuild.

A C++/FFmpeg pipeline prepares the corpus directly from competition videos by selecting one representative frame at each one-second target. The preparation job records aggregate and per-video frame counts in a generated summary and validates coverage before indexing. Caption, OCR, object, and speech stages then attach searchable evidence to the extracted timeline. Throughout preparation, `video_id`, internal `frame_id`, submission coordinate `frame_idx`, and `timestamp_ms` remain distinct. No layer derives a competition coordinate from filename order or a vector-array position.

Appendix Table 1 lists the model roles. Qwen3-VL-8B produces captions; Florence-2 reads visible text; YOLOE detects open-vocabulary objects; Qwen3-ASR-1.7B and pyannote provide timestamped speech and speakers. SigLIP2 encodes images and visual text, while BGE-M3 encodes context and ASR text [1–7]. Qwen3-4B can prepare query candidates [8], but the client must select one explicitly. An experimental VLM reranker also exists in the inference package; the current public KIS/TRAKE path does not call it, so we do not report

it as an active stage.

2. Backend Construction and Execution

At startup, the backend validates configuration and typed artifact contracts, then opens the canonical frame metadata plus available caption, OCR, object, transcript, and media artifacts. It requires the visual index and attempts to load the frame-context, ASR-segment, and BM25 bundles independently. Before admitting a source, compatibility checks compare embedding dimensions, canonical mappings, dataset identities, and model contracts. A capability map records frame-native dense, interval-native speech, and field-native lexical evidence. The serving process never prepares the corpus or builds indexes.

A unified search coordinator owns the loaded object graph. Its retrieval service binds each index to its query encoder, projection rule, cache, and fusion configuration. The temporal evidence service reuses those bindings to score complete canonical matrices. Deterministic planners and task workflows sit above the numerical services; canonical corpus resolvers and materializers control the response boundary. A separate literal-evidence view handles filtering, and media, history, or submission state cannot alter ranking artifacts. Loading the bundles once avoids reopening multi-gigabyte files per request.

Each request begins with validation and conversion into one or more ordered event queries. An explicitly selected rewrite may change encoder text, while lexical scoring and explanations retain the original. Compatible queries are batch-encoded, and independent components may score concurrently. Their results retain source identity and coverage. Detached search projects results to canonical frames before rank fusion. Temporal tasks form full-corpus scores for each event, partition them by video, decode and rank paths, and finally materialize evidence and submission coordinates from the FrameStore. Bounded embedding and thumbnail caches reduce repeated model and image work without caching mutable response objects or changing deterministic ordering.

3. Retrieval System

3.1 Canonical preparation and FrameContext

For raw video, the C++/FFmpeg sampler sets target times $T_k = k\Delta$, where $\Delta = 1000$ ms and $0 \leq T_k < D$. At

each target, it chooses the closer of the preceding and current decoded timestamps; exact ties choose the preceding frame. A video of duration D yields $\lceil D/\Delta \rceil$ samples even when decode cadence varies. The competition submission coordinate is

$$f_k = \left\lfloor \lceil \bar{r} \rceil \frac{T_k}{1000} \right\rfloor, \quad (1)$$

where $\bar{r}$ is average FPS. Extraction validates counts and identity, then atomically publishes durable 1024-pixel JPEGs and higher-quality enrichment images. The extracted records populate the canonical schema illustrated in Appendix Fig. 2.

Preparation runs in resumable, bounded batches. Before processing, the pipeline hashes the sorted source manifest and extraction configuration with SHA-256. Each batch stages its source or archive, extracts frames, runs caption/OCR/object specialists, builds FRAMECONTEXT, encodes visual and context vectors, validates or reuses timestamped ASR, and projects per-video artifacts. It commits the three-index handoff atomically before clearing scratch data. Durable images feed captioning, object detection, and visual embeddings; the higher-quality variant feeds OCR. Completion requires specialist tables to cover the expected canonical frames and pass identity checks. These checks prevent publication until each embedding has traceable text and image lineage.

Caption, OCR, and object stages retain separate specialist artifacts. A deterministic FRAMECONTEXT view serializes valid evidence as [CAPTION], [VISIBLE_TEXT], and [OBJECTS], in that order and separated by blank lines. Their whitespace-token budgets are 80, 80, and 40. Caption evidence must be complete and error-free. OCR must also be non-empty and reach quality 0.5; objects require a completed, non-empty summary. Raw OCR quality and object count remain recorded when their text is excluded. Typed fields preserve availability, OCR quality, object count, source versions, and frame-store lineage for omitted and included sections alike. The builder joins sources through bounded SQLite, checks manifests and canonical identities, and atomically publishes Parquet plus its manifest. Retaining source fields lets the system rebuild the combined text and inspect each contribution separately.

3.2 Indexes and two fusion regimes

SigLIP2 encodes every durable image into a 768-dimensional vector and BGE-M3 encodes FRAMECONTEXT and non-empty ASR segments into 1024 dimensions. All vectors are float32 and L2-normalized, so exact inner product equals cosine similarity. The frame-native bundles use FAISS IndexFlatIP; each also stores its vector matrix, canonical mapping, timestamps, and per-video posting arrays [9]. For a batch of event queries Q and corpus vectors V , full temporal scoring is the chunked matrix product

$$S^{(m)} = Q^{(m)} V^{(m)\top}. \quad (2)$$

The ASR index is segment-native: a half-open interval $[a, b)$ contributes its maximum score to covered canonical frames, with a bounded nearest-midpoint fallback only in detached top- K projection.

Each dense bundle stores the serialized index, normalized vector matrix, canonical frame or segment mappings,

metadata, timestamps, and compact per-video posting offsets/positions. Build preflight verifies FrameStore authority, expected coverage, model/index contracts, and source lineage before encoding. Online top- K search uses FAISS. Full-corpus or video-subset temporal scoring reads the normalized vector matrix and computes Eq. 2 in bounded chunks. The complete score matrix preserves candidates that shortlist alignment would discard and splits into videos in canonical order without another identity join. Queries that share an encoder family are batch-encoded; independent modalities may score concurrently.

For segment search, results first retain the strongest segment contribution per frame. Canonical frames inside $[a, b)$ are preferred; only an empty interval uses the nearest segment midpoint, and then only within the configured 5000-ms bound. Ties resolve by distance, timestamp, frame index, and frame ID. Temporal full-corpus projection is stricter: covered frames receive the maximum of all overlapping interval scores and uncovered positions carry both score zero and a false coverage mask, later consumed by Eq. 6.

A sparse index independently stores BM25 matrices for title, caption, OCR, and ASR after Unicode NFKC normalization, lowercasing, and alphanumeric tokenization. With $k_1 = 1.5$ and $b = 0.75$, a query-term contribution is

$$\begin{aligned} K_d &= k_1 \left(1 - b + b|d|/|\overline{d}| \right), \\ s_q(d) &= \text{idf}(q) \frac{tf(q, d)(k_1 + 1)}{tf(q, d) + K_d}, \\ \text{idf}(q) &= \log \left(1 + \frac{N - df(q) + 0.5}{df(q) + 0.5} \right). \end{aligned} \quad (3)$$

following the probabilistic relevance framework [10]. The current snapshot has populated title, OCR, and ASR vocabularies; its serialized BM25 caption field is empty, while semantic caption evidence remains present through FRAMECONTEXT. Appendix Table 2 records this manifest state. Sparse scoring sums the selected query-token columns instead of densifying the entire vocabulary. Original event text queries title, OCR, and ASR fields; caption-oriented retrieval text may address the caption field. Source-specific fields retain literal names, signs, and spoken phrases that dense embeddings may smooth over, and they keep synthetic combined context from being reported as a source caption.

Detached and temporal search use different fusion rules. Detached nearest-neighbor frame search combines modality lists by weighted reciprocal-rank fusion [11]:

$$R(i) = \sum_{m \in \mathcal{M}_i} \frac{w_m}{60 + \text{rank}_m(i)}. \quad (4)$$

RRF uses only rank over the union of returned frame IDs and preserves source provenance. One-based ranks, active-weight normalization, a required visual source, and deterministic tie-breaking by fused score, best source rank, then frame ID make repeated queries stable. ASR segments are projected before fusion, so every list member already has a canonical frame identity. The KIS/TRAKE temporal scorer operates on calibrated full-corpus scores.

Temporal search scores every event against every canonical frame. For component m and event e , supported

values are quantile-clipped and min-max mapped to $\hat{S}_{met} \in [0, 1]$. Dense reliability measures upper-tail contrast:

$$z_{me} = \frac{\max(\mu_{me}^{\text{top}} - \text{med}_{me}, 0)}{\text{IQR}_{me} + \varepsilon}, \quad \rho_{me} = \frac{z_{me}}{1 + z_{me}}. \quad (5)$$

A lexical row receives binary reliability when any exact match exists. A deterministic cue router adjusts base weights for speech, visible text, or visual language. Coverage $c_{met} \in \{0, 1\}$ excludes absent segment or evidence data from the fused score:

$$F_{et} = \frac{\sum_m w_{me} \rho_{me} c_{met} \hat{S}_{met}}{\sum_m w_{me} \rho_{me} c_{met}}, \quad w_{me} = w_m^0 g_m(q_e). \quad (6)$$

When no nonvisual component covers a frame, calibrated visual evidence supplies the fallback. Calibration uses the 5th percentile and maximum as bounds, the strongest one percent capped at 128 frames for μ^{top} , and $\varepsilon = 10^{-6}$. Base masses are 0.35 visual, 0.35 context, 0.08 dense ASR, and 0.02/0.10/0.04/0.06 for title/caption/OCR/ASR BM25. Cue multipliers are 1.4 for visual language, 5.0 for speech, and 3.0 for visible text. Eq. 6 renormalizes positive mass at every frame. These values are deterministic configuration policies, not learned weights. Per-frame renormalization distinguishes missing evidence from weak evidence and yields one fused event-frame matrix per video.

3.3 KIS/TRAKE temporal orchestration

KIS converts a natural-language description into ordered moments with a deterministic Vietnamese-aware planner. An explicitly selected query-preparation candidate may replace only retrieval text; BM25 and explanations retain the original. TRAKE accepts ordered events directly. Both invoke the same temporal service to build component matrices, calibrate and fuse them, split canonical positions by video, decode paths, and revalidate identities before response materialization (Appendix Fig. 4).

The planner prefers non-empty lines, then sentence boundaries. It removes a trailing question, folds attribute or enumeration sentences into the preceding moment, and recognizes Vietnamese and English transition cues. A “before that” moment moves before its referent; later cues preserve forward order. Deterministic planning lets UI replay and lexical evidence display the exact aligned text. Request validation caps the sequence at 32 events and rejects an empty sequence before allocating a full-corpus matrix.

For one video, let F_{et} be the fused score for event e at frame t and τ_t its timestamp. Strict chronological alignment uses

$$D_{0t} = F_{0t}, \quad D_{et} = F_{et} + \max_{u < t} [D_{e-1,u} - \lambda(\tau_t - \tau_u)]. \quad (7)$$

The gap penalty ($\lambda = 10^{-5}$ by default) discourages unnecessarily distant events without imposing a fixed window. Rewriting the predecessor term as

$$D_{et} = F_{et} - \lambda\tau_t + \max_{u < t} (D_{e-1,u} + \lambda\tau_u) \quad (8)$$

permits a prefix-maximum and backpointer scan in $O(ET)$ time and $O(ET)$ backpointer space instead of

$O(ET^2)$. At each position, cumulative maxima store the best shifted predecessor value and its canonical position; equal values keep the latest position. Endpoints with non-finite state are unreachable, and a video with fewer admissible frames or score clusters than events returns no path. An optional score exponent raises non-negative values before decoding; the default 1.0 leaves scores unchanged. Optional score-profile clusters keep multiple events out of a near-identical region. The runtime retains up to five alternatives per video and suppresses final-event alternatives within 30 seconds before global ranking.

KIS uses the upper-middle aligned event, $\lfloor |P|/2 \rfloor$, as its representative result and retains the complete path for explanation and navigation. TRAKE returns every ordered canonical frame/timestamp sequence and submits the entire path as one row. Before materialization, the shared temporal service resolves every DP position through the FrameStore and verifies its video ID, frame ID, frame index, and timestamp. KIS attaches inspectable evidence for the representative frame plus the full alignment; TRAKE preserves every returned path array unchanged. Responses expose competition-facing coordinates and evidence, while optimizer array offsets remain internal.

4. Architectural Evolution

Version 1 kept temporal alignment outside the KIS and question-style retrieval paths. Each path ran dense nearest-neighbor search over visual frames, FRAMECONTEXT, and ASR segments, then applied shallow alignment to the returned candidates. Sequence reasoning saw only this shortlist. It could not recover a frame that scored weakly for an isolated event but completed a coherent multi-event path, and separate task paths could encode different temporal assumptions.

Version 2 routes every competition retrieval workflow through the shared temporal alignment pipeline. Once input becomes ordered events, the pipeline scores complete modality matrices, calibrates and fuses covered evidence, partitions canonical positions by video, and runs monotonic DP before final truncation. KIS and TRAKE now use the same optimizer and canonical backpointers. KIS accepts natural-language input and presents one representative frame while retaining its path; TRAKE accepts explicit ordered events, then presents and submits the complete path. Detached dense retrieval remains an inspection utility. The team observed a substantial qualitative improvement in practical multi-event retrieval after this change. Quantitative evaluation remains outside this report.

5. Representative Query Analysis

5.1 Map and four hydraulic-structure appearances

Query: “Find the clip that begins with a map, after which one type of hydraulic structure appears four times in sequence.”

This query combines an opening anchor, event order, same-video continuity, and an exact recurrence count. In v1, the team searched for each scene independently and merged the Top- K lists. Returned frames often came from different videos or appeared out of order. A weak intermediate occurrence could also fall outside its shortlist and break the entire sequence.

In v2, the team encodes the description as ordered events. The temporal service scores each event over the full timeline, partitions frames by video, and runs monotonic DP before selecting Top- K paths. The returned paths preserve the requested progression instead of matching isolated frames. The result view shows the complete alignment and opens each moment in the video inspector, allowing members to verify the sequence against the timeline.

The exact-four condition still needs manual review because the evidence pipeline does not count intermittent reappearances reliably. Team members must inspect the returned path and video timeline to confirm all four occurrences. Temporal DP enforces order and same-video continuity, but it does not solve cardinality.

5.2 Cycling formation, street name, and countdown

Query: “Find a rear-view shot of the lead group, with one rider ahead and three following, as they turn right onto Ho Tung Mau Street at an intersection whose green signal is counting down to 13 seconds.”

The rider formation and turn are visual-semantic cues; the street name and the number 13 are decisive literal cues. Dense retrieval matched broad concepts such as a cycling group and an intersection, but the proper noun and numeral were diluted in the embedding. The team combined dense evidence with fielded BM25 over title, OCR, and ASR. These fields can promote rare terms from road signs and speech, while the inspector exposes their source evidence to the operator.

The case still failed. ASR mistranscribed “Ho Tung Mau,” and OCR did not extract 13, leaving BM25 without either decisive token. The visible OCR and ASR fields helped the team diagnose the miss but could not recover the absent evidence. BM25 can rank only terms present in its index; it cannot repair upstream OCR or ASR errors. Evidence extraction was the limiting stage for this query.

6. Operator Interface (Brief)

The interface maps plain text to KIS and explicit E1:, E2:, ... lines to TRAKE. Operators select Top- K and dense or lexical families, inspect result cards and complete alignments, then open video at the retrieved timestamp. Caption, OCR, object, ASR, and canonical-coordinate evidence appear beside the video. Appendix Figs. 6–8 show these interactions.

The collaborative workspace stores user-scoped query replay, manual video/timestamp opening, and shared CSV answer files (Fig. 9). SQLite records revisions and committed rows; synchronization distributes revision-aware file commands. KIS appends one video/frame pair, while TRAKE appends a complete ordered path as one row. The UI supports editing, validation, and ZIP export. Chat, shared cursors, and presence are outside the implemented workspace.

7. Thunder Compute Execution Boundary

The index-build runbook assigns one RTX A6000 ephemeral worker. It inventories or downloads only canonical frames, keyframes, FRAMECONTEXT, and transcripts from S3; builds visual, context, and ASR-segment bundles; validates coverage and identity; and publishes a checksummed version after its build report passes. For

online inference, an operator can create a disposable GPU VM, copy the selected model package, start the inference API, and expose a protected Cloudflare endpoint. The GPU SKU remains an operational choice: the checked-in inference example uses an L40. Instance creation, upload, health checks, and deletion are manual operator steps. VM teardown leaves local FastAPI state and S3 artifacts intact [12].

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A. Architecture and Component Map

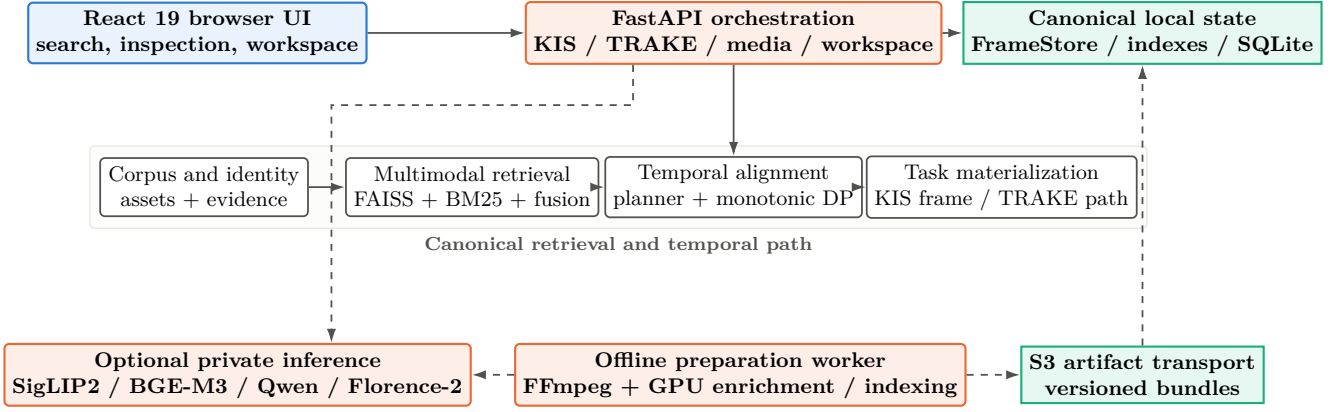


Figure 1: System architecture and execution boundaries. Solid arrows show online control and data flow; dashed arrows show optional inference or artifact transport. Canonical retrieval state stays local, and the GPU service is replaceable. Labels and borders duplicate every color cue.

Table 1: Model and algorithm inventory. Status marks each component as active in the offline or online path, configurable, optional, or not wired.

Capability	Model / algorithm	Produced evidence and downstream use	Status
Caption	Qwen/Qwen3-VL-8B-Instruct	Frame caption; first FRAMECONTEXT section and inspectable specialist artifact	Offline active
Visible text	Florence-2-base-ft	OCR regions, frame summary, quality; gated FRAMECONTEXT section and BM25 OCR field	Offline active
Objects	YOLOE-26l-seg	Detections and normalized object summary; FRAMECONTEXT section and literal filter	Offline active
Speech	Qwen3-ASR-1.7B-hf	Timestamped transcript segments; BGE-M3 segment index and BM25 ASR field	Offline active
Speakers	pyannote community-1	Diarization attached to transcript segments	Configurable
Visual encoder	SigLIP2 base patch16-224	768-D image vectors and visual-language query vectors	Active
Text encoder	BAAI/bge-m3	1024-D FRAMECONTEXT and ASR-segment vectors	Active
Dense search	FAISS IndexFlatIP	Exact inner product over L2-normalized vectors; cosine-equivalent ranking	Active
Lexical search	Fielded BM25	Title/caption/OCR/ASR literal evidence	Active
Query preparation	Qwen/Qwen3-4B	Five optional candidate rewrites selected explicitly by the client	Optional
Temporal decoding	Prefix-max monotonic DP	Strict same-video event paths, alternatives, and canonical backpointers	KIS/TRAKE active
VLM reranking	Qwen3-VL reranker adapter	Experimental package capability; absent from public request contract and UI	Not wired

B. Offline Preparation and Index Construction

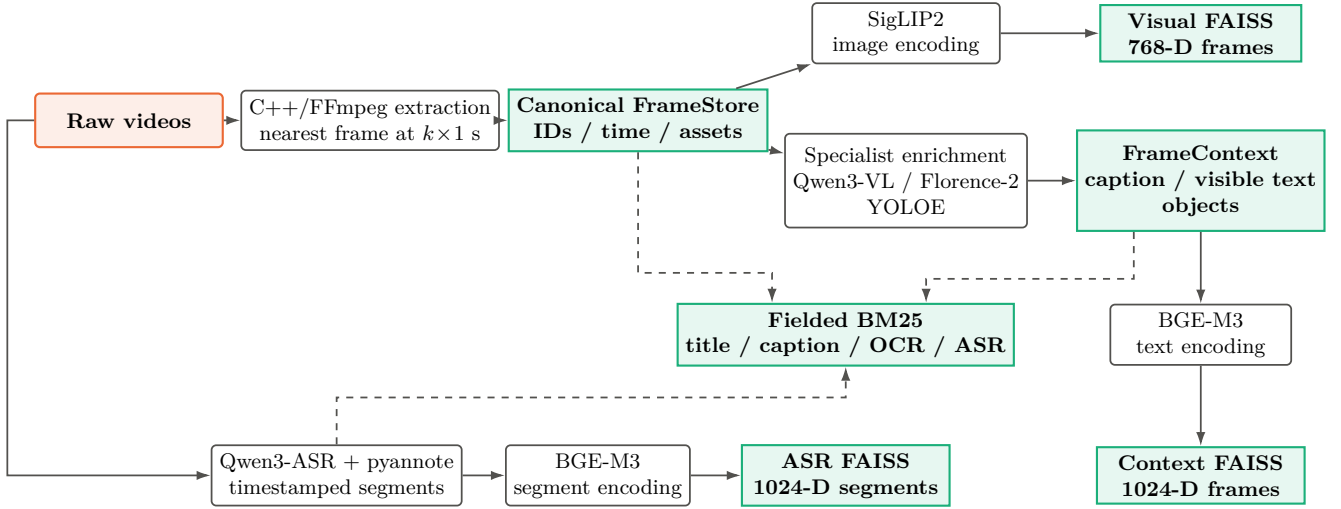


Figure 2: Offline preparation from C++/FFmpeg self-extracted 1-FPS frames. Solid arrows show model and encoding stages; dashed arrows feed the lexical index. Visual and context indexes remain frame-native, while the ASR index remains segment-native.

B.1 Corpus Statistics

The reporting tool behind Fig. 3 scans one lightweight identity column at a time in batches of at most 8,192 rows. It excludes intermediate batch shards and never loads a complete Parquet table. Before rendering the chart, it checks streamed totals against Parquet footers and the frame-preparation manifest.

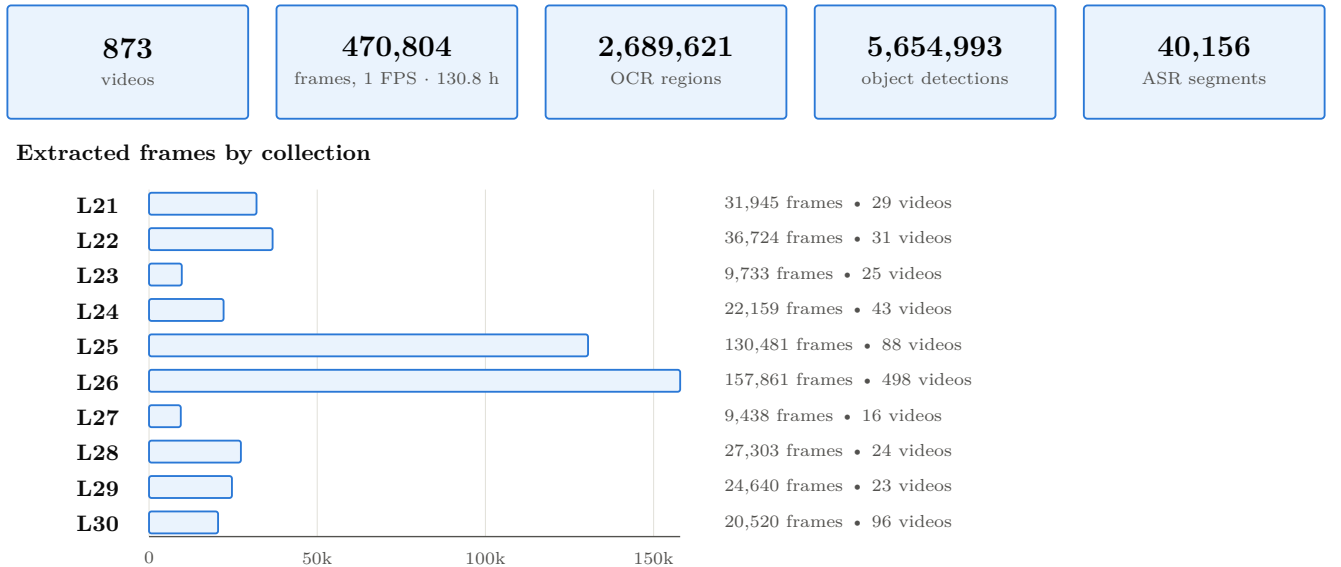


Figure 3: Corpus summary streamed in bounded batches from finalized artifacts. All bars share a zero-based frame-count axis and include exact frame and video labels. At one frame per second, 470,804 frames span approximately 130.8 sampled timeline hours.

Table 2: Generated corpus and index counts. They document system scale, not retrieval effectiveness.

Artifact	Entities	Representation	Observed contract / caveat
Extracted frame corpus	873 videos; 470,804 frames	C++/FFmpeg sampling at 1 FPS	Per-video and aggregate frame counts validated before indexing
Caption / OCR / objects / context	470,804 frame-aligned rows per artifact	Specialist Parquet + deterministic FRAMECONTEXT	2,689,621 OCR regions and 5,654,993 object detections
Visual index	470,804 frames	768-D float32, L2, flat IP	Model name recorded; serialized model revision/source/config fingerprints absent
Vietnamese context index	470,804 frames	1024-D float32, L2, flat IP	Frame-aligned context vectors; revision/retrieval-source fingerprints remain to be finalized
ASR segment index	40,156 intervals	1024-D float32, L2, flat IP	BGE-M3 revision recorded; half-open temporal geometry retained
BM25 index	470,804 documents	Four sparse field matrices	$k_1 = 1.5$, $b = 0.75$; current caption vocabulary size is zero

Table 3: Deterministic retrieval policies and defaults.

Stage	Parameter	Policy
Sampling	$\Delta = 1000$ ms	Select nearest decoded timestamp; ties choose preceding frame; targets lie in $[0, D)$
FrameContext	budgets 80/80/40; OCR ≥ 0.5	Ordered caption, visible-text, object sections; preserve lineage and availability
Dense bundles	float32, L2, flat IP	Exact cosine-equivalent search; vectors, mappings, postings, and timestamps stored together
ASR projection	maximum gap 5000 ms	Prefer frames inside $[a, b)$; otherwise nearest midpoint within bound for detached search
Detached fusion	RRF constant 60	Weighted union of modality top- K lists; visual source required
Temporal fusion	adaptive, coverage-aware	Quantile calibration, reliability gating, cue routing, per-frame renormalization
Alignment	$\lambda = 10^{-5}$; power 1	Strict monotonic path; prefix-max recurrence; latest predecessor on ties
Alternatives	5 paths/video; 30 s separation	Suppress near-duplicate final-event moments before global ranking

C. Online Fusion and Temporal Task Orchestration

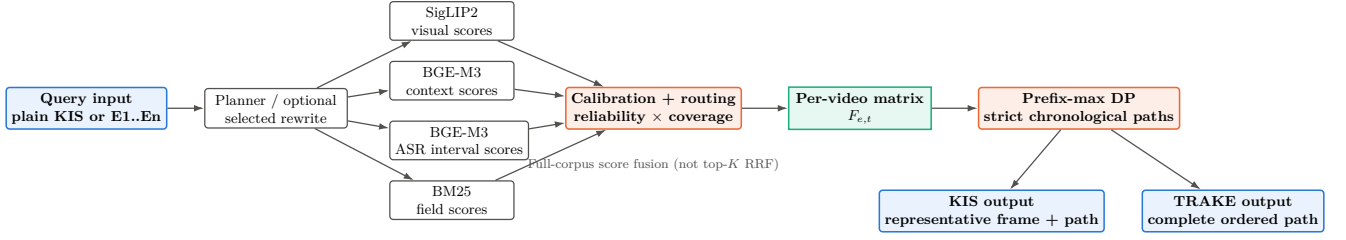


Figure 4: Shared KIS/TRAKE temporal path. Explicit rewrites leave original text available to lexical scoring. Reliability and coverage modify component scores before monotonic decoding and task-specific materialization.

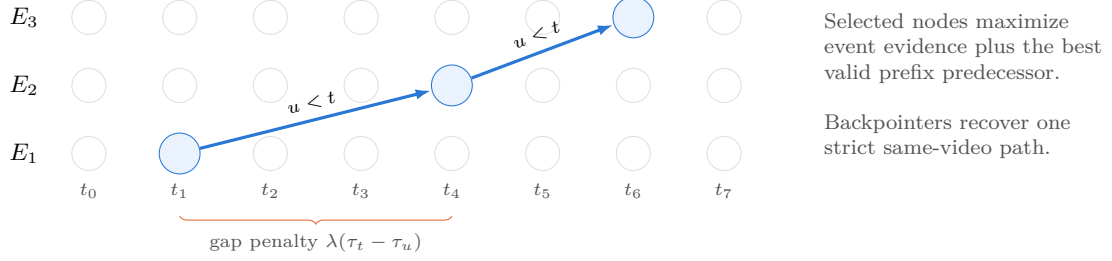


Figure 5: Monotonic dynamic programming. Each event selects a strictly later canonical frame. Prefix maxima replace all-pairs predecessor search, and backpointers retain exact frame identities.

D. Live Operator Interface

These 1440×900 screenshots come from the running React/FastAPI application. An isolated, identity-preserving capture profile loaded 21 videos and 3,807 frames from the documented corpus, together with their original visual vectors and specialist evidence; only visual dense retrieval was enabled. Because the original L30_V060 video was unavailable locally, the inspector uses a playback proxy assembled from its canonical 1-FPS frames. The proxy exercises the real range-streaming and timeline code but is not the source video. Corpus counts and algorithm descriptions elsewhere refer to the full artifact manifests, not this fixture. Numbered callouts and component borders duplicate color cues.

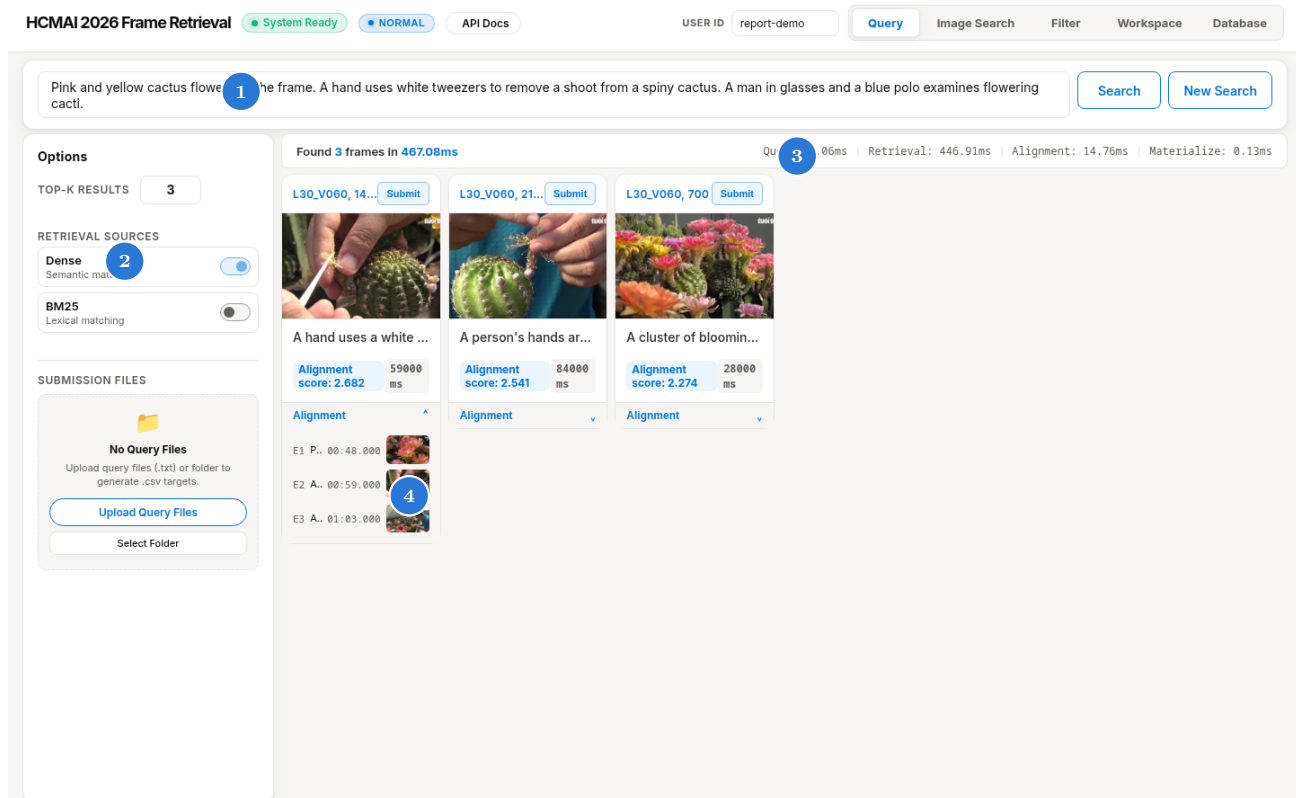


Figure 6: KIS retrieval workspace: [1] natural-language query; [2] Top- K and dense/BM25 controls beside shared answer files; [3] latency and result summary; [4] canonical result cards with an expanded event alignment and submission action.

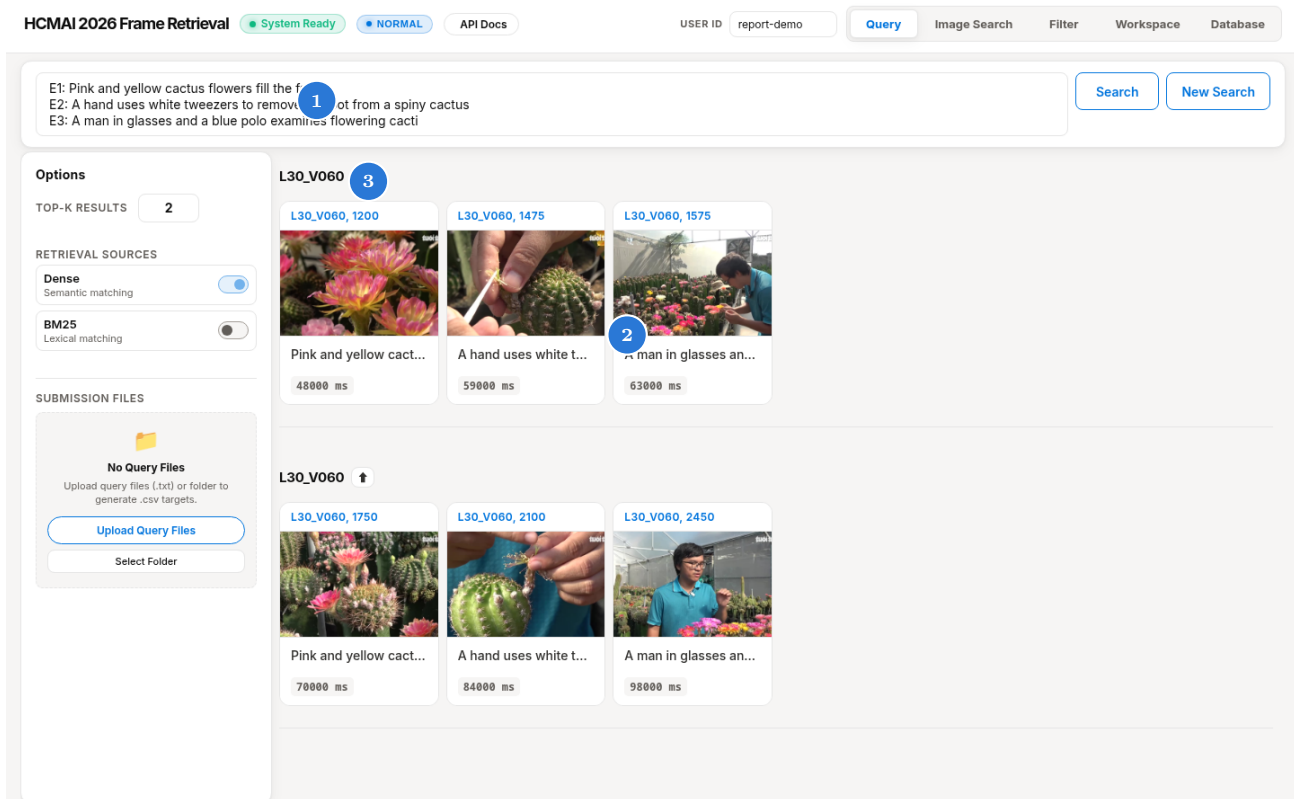


Figure 7: TRAKE ordered-event retrieval: [1] sequential E1..En input; [2] same-video path in event order; [3] path-level action that writes one ordered multi-frame CSV row.

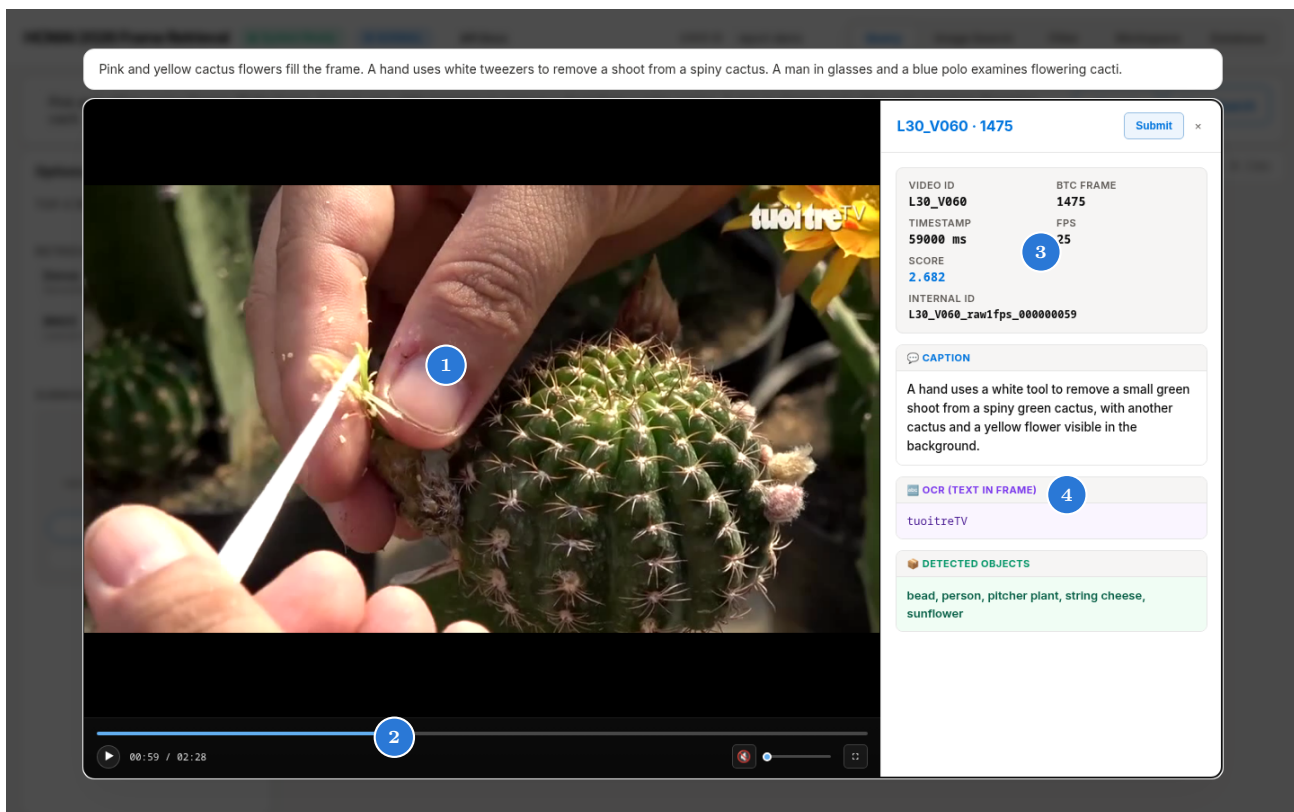


Figure 8: Video evidence inspector: [1] preview at the retrieved moment; [2] custom timeline and precise seek controls; [3] live canonical and submission coordinates; [4] available caption, OCR, object, and other evidence for human verification.

HCMAI 2026 Frame Retrieval System Ready NORMAL API Docs USER ID: report-demo Query Image Search Filter Workspace Database

History

E1: Pink and yellow cactus flowers fill the frame E2: A hand uses white tweezers to remove a shoot from a spiny cactus E3: A man in glasses and a blue polo examines flowering cacti

query-demo-trake.csv

1

Replay in Query

Pink and yellow cactus flowers fill the frame. A hand uses white tweezers to remove a shoot from a spiny cactus. A man in glasses and a blue polo examines flowering cacti.

query-demo-kis.csv

Replay in Query

E1: Pink and yellow cactus flowers fill the frame E2: A hand uses white tweezers to remove a shoot from a spiny cactus E3: A man in glasses and a blue polo examines flowering cacti

Replay in Query

Pink and yellow cactus flowers fill the frame. A hand uses white tweezers to remove a shoot from a spiny cactus. A man in glasses and a blue polo examines flowering cacti.

Replay in Query

Pink and yellow cactus flowers fill the frame. A hand uses white tweezers to remove a shoot from a spiny cactus. A man in glasses and a blue polo examines flowering cacti.

Replay in Query

Close-ups of colorful cactus flowers in a greenhouse. A person uses a small hand tool on a cactus. A man presents a flowering cactus.

Replay in Query

A person tends blooming cacti with a small hand tool inside a greenhouse

Replay in Query

Open video moment

VIDEO_ID: L30_V060 2 TIMESTAMP_MS: 59000

Open in viewer

SUBMISSION FILES

query-demo-kis.csv 3

query-demo-trake.csv

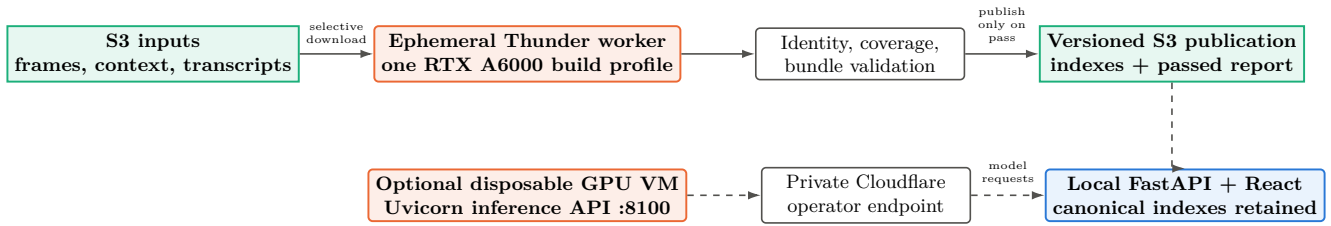
Download 4 SV ZIP (2)

Figure 9: Collaborative workspace: [1] user-scoped query history and replay; [2] manual video/timestamp opening; [3] shared SQLite-backed submission files with distinct validated and filled states; [4] multi-file CSV ZIP export.

E. Traceability and Deployment Evidence

Table 4: Traceability from body subsystems to appendix evidence and technique boundaries.

Presented subsystem	Appendix evidence	Technique / method boundary
End-to-end architecture / services	Fig. 1; Table 1	Orchestration, task routing, and model coordination
Keyframe ingestion / identity	Fig. 2; Table 3	C++/FFmpeg 1-FPS sampling, canonical identity schema
Caption, OCR, objects, ASR, FrameContext	Fig. 2; Tables 1–2	Specialist enrichment and deterministic context serialization
FAISS / BM25 / fusion	Figs. 2 and 4; Tables 2–3	Dense/sparse indexing, RRF and adaptive evidence fusion
Temporal DP / KIS / TRAKE	Figs. 4 and 5	Prefix-max monotonic DP, calibration, cue routing
KIS operator flow	Fig. 6	Natural-language planner, representative frame selection
TRAKE operator flow	Fig. 7	Ordered-event input, complete same-video path submission
Video preview / evidence inspection	Fig. 8	Frame-accurate seek, specialist evidence overlay
Collaborative workspace / submissions	Fig. 9	Revision-aware shared answer files, query history replay
Thunder Compute / S3 execution	Fig. 1; text below	Ephemeral GPU build, versioned artifact publication



Manual lifecycle: create → upload → health check → delete.

Figure 10: Separate Thunder Compute lifecycles for index construction and optional online inference. Versioned serving artifacts remain outside the VM and survive instance deletion.

Release note. The runbook requires a passing build report before publication. The local bundle in Table 2 lacks the promoted build report and complete visual/context provenance fields; an official release must reconcile both. The inference example uses an L40, while the index-build profile specifies an A6000. Figure 10 labels each GPU by workflow.